

# Cassandra Lab 2 Part 2

## 1.Create Keyspace: EmployeeDB

bmscecse@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~\$ cqlsh

Connected to Test Cluster at [127.0.0.1:9042](http://127.0.0.1:9042)

[cqlsh 6.1.0 | Cassandra 4.1.4 | CQL spec 3.4.6 | Native protocol v5]

Use HELP for help.

cqlsh> CREATE KEYSPACE IF NOT EXISTS EmployeeDB

```
... WITH REPLICATION = {  
...   'class' : 'SimpleStrategy',  
...   'replication_factor' : 1  
... };
```

cqlsh> DESCRIBE KEYSPACES;

```
education  library  students  system_distributed  system_views  
employee   student  system    system_schema      system_virtual_schema  
employeeadb student1  system_auth  system_traces
```

cqlsh> USE EmployeeDB;

## 2.Create Table: Employee\_Info

cqlsh:employeeadb> CREATE TABLE IF NOT EXISTS Employee\_Info (

```
...   Emp_Id INT PRIMARY KEY,  
...   Emp_Name TEXT,  
...   Designation TEXT,  
...   Date_of_Joining DATE,  
...   Salary INT,  
...   Dept_Name TEXT,  
...   Projects SET<TEXT>  
... );
```

cqlsh:employeeadb> DESCRIBE TABLES;

employee\_info

## Cassandra Lab 2 Part 2

### 3.Insert Values Using BATCH

```
cqlsh:employee> BEGIN BATCH
...
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining,
Salary, Dept_Name)
... VALUES (121, 'Amit', 'Software Engineer', '2019-06-15', 60000, 'IT');
...
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining,
Salary, Dept_Name)
... VALUES (122, 'Ravi', 'Senior Engineer', '2018-03-20', 75000, 'R&D');
...
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining,
Salary, Dept_Name)
... VALUES (123, 'Meera', 'Manager', '2016-01-10', 90000, 'HR');
...
... APPLY BATCH;
cqlsh:employee> SELECT * FROM Employee_Info;
```

| emp_id | date_of_joining | dept_name | designation       | emp_name | projects | salary |
|--------|-----------------|-----------|-------------------|----------|----------|--------|
| 123    | 2016-01-10      | HR        | Manager           | Meera    | null     | 90000  |
| 122    | 2018-03-20      | R&D       | Senior Engineer   | Ravi     | null     | 75000  |
| 121    | 2019-06-15      | IT        | Software Engineer | Amit     | null     | 60000  |

(3 rows)

### 4.Update Employee Name and Department (Emp\_Id = 121)

```
cqlsh:employee> UPDATE Employee_Info
... SET Emp_Name = 'Amit Kumar',
... Dept_Name = 'Data Science'
... WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info;
```

| emp_id | date_of_joining | dept_name    | designation       | emp_name   | projects | salary |
|--------|-----------------|--------------|-------------------|------------|----------|--------|
| 123    | 2016-01-10      | HR           | Manager           | Meera      | null     | 90000  |
| 122    | 2018-03-20      | R&D          | Senior Engineer   | Ravi       | null     | 75000  |
| 121    | 2019-06-15      | Data Science | Software Engineer | Amit Kumar | null     | 60000  |

(3 rows)

## Cassandra Lab 2 Part 2

### 5.Sort Employee Records Based on Salary

```
cqlsh:employee> CREATE TABLE IF NOT EXISTS Employee_By_Salary (
...   Dept_Name TEXT,
...   Salary INT,
...   Emp_Id INT,
...   Emp_Name TEXT,
...   PRIMARY KEY (Dept_Name, Salary, Emp_Id)
... ) WITH CLUSTERING ORDER BY (Salary DESC);
cqlsh:employee> INSERT INTO Employee_By_Salary (Dept_Name, Salary, Emp_Id,
Emp_Name)
... VALUES ('IT', 60000, 121, 'Amit Kumar');
cqlsh:employee>
cqlsh:employee> INSERT INTO Employee_By_Salary (Dept_Name, Salary, Emp_Id,
Emp_Name)
... VALUES ('IT', 75000, 122, 'Ravi');
cqlsh:employee>
cqlsh:employee> INSERT INTO Employee_By_Salary (Dept_Name, Salary, Emp_Id,
Emp_Name)
... VALUES ('IT', 50000, 123, 'Meera');
cqlsh:employee> SELECT * FROM Employee_By_Salary
... WHERE Dept_Name = 'IT';
```

```
dept_name | salary | emp_id | emp_name
-----+-----+-----+-----
IT | 75000 | 122 | Ravi
IT | 60000 | 121 | Amit Kumar
IT | 50000 | 123 | Meera
```

(3 rows)

## Cassandra Lab 2 Part 2

### 6.Alter Table to Add Projects Column

```
cqlsh:employee> ALTER TABLE Employee_Info
... ADD Projects SET<TEXT>;
cqlsh:employee> SELECT * FROM Employee_Info;
```

| emp_id | date_of_joining | dept_name    | designation       | emp_name   | projects | salary |
|--------|-----------------|--------------|-------------------|------------|----------|--------|
| 123    | 2016-01-10      | HR           | Manager           | Meera      | null     | 90000  |
| 122    | 2018-03-20      | R&D          | Senior Engineer   | Ravi       | null     | 75000  |
| 121    | 2019-06-15      | Data Science | Software Engineer | Amit Kumar | null     | 60000  |

### 7.Update Projects Column

```
UPDATE Employee_Info
SET Projects = {'AI-Analytics', 'Cloud-Migration', 'BigData-Pipeline'}
WHERE Emp_Id = 121;
```

```
UPDATE Employee_Info
SET Projects = Projects + {'ML-Deployment'}
WHERE Emp_Id = 121;
```

```
SELECT Emp_Id, Projects
FROM Employee_Info
WHERE Emp_Id = 121;
```

(1 rows)

```
cqlsh:employee> SELECT * FROM Employee_Info;
```

| emp_id | date_of_joining | dept_name    | designation       | emp_name   | projects                                                                 | salary |
|--------|-----------------|--------------|-------------------|------------|--------------------------------------------------------------------------|--------|
| 123    | 2016-01-10      | HR           | Manager           | Meera      |                                                                          | 90000  |
| 122    | 2018-03-20      | R&D          | Senior Engineer   | Ravi       |                                                                          | 75000  |
| 121    | 2019-06-15      | Data Science | Software Engineer | Amit Kumar | {'AI-Analytics', 'BigData-Pipeline', 'Cloud-Migration', 'ML-Deployment'} | 60000  |

(3 rows)

## Cassandra Lab 2 Part 2

### 8. Insert Record with TTL (15 Seconds)

```
cqlsh:employeeedb> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation,  
Date_of_Joining, Salary, Dept_Name)
```

```
... VALUES (130, 'TemporaryUser', 'Intern', '2024-01-01', 15000, 'Training')
```

```
... USING TTL 15;
```

```
cqlsh:employeeedb> SELECT * FROM Employee_Info WHERE Emp_Id = 130;
```

| emp_id | date_of_joining | dept_name | designation | emp_name      | projects | salary |
|--------|-----------------|-----------|-------------|---------------|----------|--------|
| 130    | 2024-01-01      | Training  | Intern      | TemporaryUser | null     | 15000  |